

Smoke Point in Co-flow Experiment (SPICE)

Laminar Smoke Point Properties of Non-buoyant Round-Jet Diffusion Flames

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Abstract

The laminar smoke point is the condition (and the luminous flame length) where a flame begins to emit soot. Smoke points indicate the sooting propensity of a fuel in diffusion flames. In normal gravity, laminar smoke points have been measured for many gaseous and condensed fuels. Laminar smoke points have been found to correlate with soot volume fractions and radiative loss fractions of turbulent diffusion flames. SPICE project measured laminar smoke points in non-buoyant laminar jet diffusion flames in co-flowing air aboard the International Space Station. Fifty-two smoke points were found for ethylene, propane, propylene, and propylene/nitrogen mixtures. Burner diameters were 0.41, 0.76, and 1.6 mm and co-flow velocities varied from 5.4 – 65 cm/s. These flames afford extensive control over residence time via variations in dilution, burner diameter, and co-flow velocity. Increasing co-flow velocity decreases luminous flame lengths, but moves smoke points to higher fuel flow rates and longer luminous flame lengths. Smaller burners have longer smoke points for co-flowing flames. These trends can be explained qualitatively by considering total residence time effects. The luminous flame lengths generally correlate with fuel mass flow rate divided by stoichiometric mixture fraction, although they increase slightly as air velocity decreases. The measured smoke point lengths were found to correlate with burner diameter to the -0.91 power times coflow velocity to the 0.41 power. The measurements allow estimates of pseudo threshold soot indices (TSI) for microgravity diffusion flames, indicating an unusually high sooting propensity for propane in microgravity. A model of residence times in coflowing flames is developed, and although residence time helps explain many of the observations, it does not correlate the measured smoke points. The measurements yielded estimates of sooting propensities of the present fuels in microgravity diffusion flames.

Acknowledgments

The authors acknowledge G.M. Faeth for conceiving this work, the SPICE engineering team, and the astronauts that conducted the tests (M. Barratt, E.M. Fincke, S.H. Magnus, and K. Wakata). The SPICE project was conducted through the [ISS Exploration Research Project of the Exploration Technology Development Program](#). The support of this program is also acknowledged.

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Nomenclature

d	= jet exit diameter
L_{SP}	= smoke point luminous flame length
$\dot{m}_f$	= fuel mass flow rate
Re	= jet Reynolds number, $4\dot{m} / (\pi d \mu)$
T_{ad}	= adiabatic flame temperature.
u_a	= streamwise velocity of co-flow air
u_f	= streamwise velocity of fuel gas.
Z_{st}	= stoichiometric mixture fraction
ϕ	= fuel-equivalence ratios.

Acronyms

ISS	International Space Station
MSG	Microgravity Science Glovebox
STS	Space Transportation System
GRC	NASA John Glenn Research Center
LSP	Laminar Soot Processes experiment
KSC	NASA John Kennedy Research Center
ULF-2	An ISS assembly mission carried out by Space Shuttle Endeavour on Nov. 14, 2008.

INTRODUCTION

SPICE project, from it was awarded in 1997 to it was conducted onboard ISS in 2009, covered more than a decade time span. During these years, the project experienced a few cycles of active and inactive status. A brief project history is given below. SPICE project was awarded on December 3, 1997 by the Microgravity Science and Application Division of NASA Headquarters to Professor Gerard M. Faeth of the University of Michigan for his proposal titled as Laminar Smoke Point Properties of Nonbuoyant Round-Jet Diffusion Flames in response to NASA's 1996 Solicitation for Glovebox Investigations issued in late 1996. The original proposal of Professor Faeth was meant to be a supplement to the Laminar Soot Processes Experiment which was a combustion project under NASA Grant No. NAG3-1245, conducted onboard STS-73 and STS-94 in 1997. In the awarding letter, the experiment was planned to be carried out onboard STS-107, for which the launch date was then scheduled in May 2000. From December 1997 to the end of 1999 was the first phase of the project. In this period, the experiment was planned to be conducted in the Middle Deck Glove Box of a Space Shuttle and the SPICE hardware was fabricated and assembled in GRC. Various project reviews necessary for the flight project, were completed. In addition, significant efforts on science side were made to explore proper camera settings for different flame conditions by conducting reduced gravity experiment in parabolic flight on board the KC-135 aircraft. From the end of 1999 to March 2002, the project was inactive. The second phase of the project was from March 2002 to January 2005. In this period, SPICE was a part of MRDOC contract to ZIN Inc., for a potential UF-1 Flight. Since the target facility of SPICE was switched from Space Shuttle's Middle Deck Glove Box to the ISS Microgravity Science Glovebox (MSG), ZIN's major effort was to retro-fit hardware for use with MSG in International Space Station (ISS), which included: designed and assembled Power/Video Unit to support MSG video interfaces, designed and fabricated a new mechanical camera supports, redesigned fuel bottle assemblies and developed new cabling. Project was cancelled again at the end of 2004. ZIN developed storage plan for hardware which was presented to NASA in December, 2004. The SPICE apparatus was put in storage in January 2005 and all the documents were archived waiting for future flight opportunities. At that time, the LSP project and the re-flight LSP-2 were completed, thus the objectives of SPICE were later changed to a stand-alone investigation of smoke point of jet diffusion flames with coexistence of air flows. One important event soon after the SPICE became inactive again was the sudden death of Professor Faeth of the University of Michigan, the PI of the project. It was not until March 2006 that the SPICE was resurrected in May, 2006 as part of the Small Payloads Projects Task under MRDOC that Dr. David Urban, the Co-I of the project, succeeded Professor Faeth as the PI of the project. In addition, Professor Peter Sunderland of University of Maryland, former Ph.D. student of Prof. Faeth was recruited as a Co-I of the project. An updated Science Requirement Document was edited. The test matrix was finalized. Additional tests in camera were conducted.

Minor hardware modifications were made and a number of project reviews were conducted. The SPICE hardware was shipped to KSC on June 18, 2008 and later lifted up to ISS by ULF-2 in October 2008. SPICE experiments were finally conducted in ISS MSG from February through June of 2009. What Follows is the science significance and justification of the SPICE project.

Soot formation in hydrocarbon-fueled flames is an important unresolved problem in combustion science. Soot is estimated to be responsible for up to 52,000 premature deaths annually in the U.S.¹ and is believed to contribute to climate change and glacial melting.² Radiation from soot in flames is beneficial in many applications but also contributes to unwanted heat loads in engines and the spread of fires. Carbon monoxide emitted by smoking flames is the major cause of death in fires.³ Knowledge of Soot formation is needed for developing computational combustion.

The differences between soot processes in buoyant and nonbuoyant laminar jet diffusion flames can be seen by comparing their soot production regions as plotted in Fig. 1. Soot formation in diffusion flames generally is limited to fuel-equivalence ratios, ϕ , in the range 1-2, which are marked on the plots. Unlike gases, which both convect and diffuse, soot mainly convects with the flow velocity so that it moves along the streamline. For buoyant flames, the dividing streamline generally is inside the soot formation region so that most of the soot initially forms near the flame sheet, where $\phi = 1$, and then moves radially inward toward cooler regions having higher fuel concentrations. The soot formation process for nonbuoyant flames is completely different: most of the soot initially forms near the cool core of the flame, where ϕ is close to 2, and is drawn directly toward the flame sheet. Results from an MSL-1 combustion project, the Laminar Soot Processes (LSP), revealed that the smoke height phenomenon in microgravity can be substantially different from that seen in normal gravity. The removal of buoyancy induced acceleration can cause the smoke point phenomenon to be dominated by radiative heat losses for conditions of low overall flame velocity. Under these conditions the flame extinguishes at the flame tip allowing soot to be emitted. In spite of this effect, it is still possible to have flames that either emit soot along stream lines or do not emit soot anywhere, for conditions where radiation losses do not dominate.

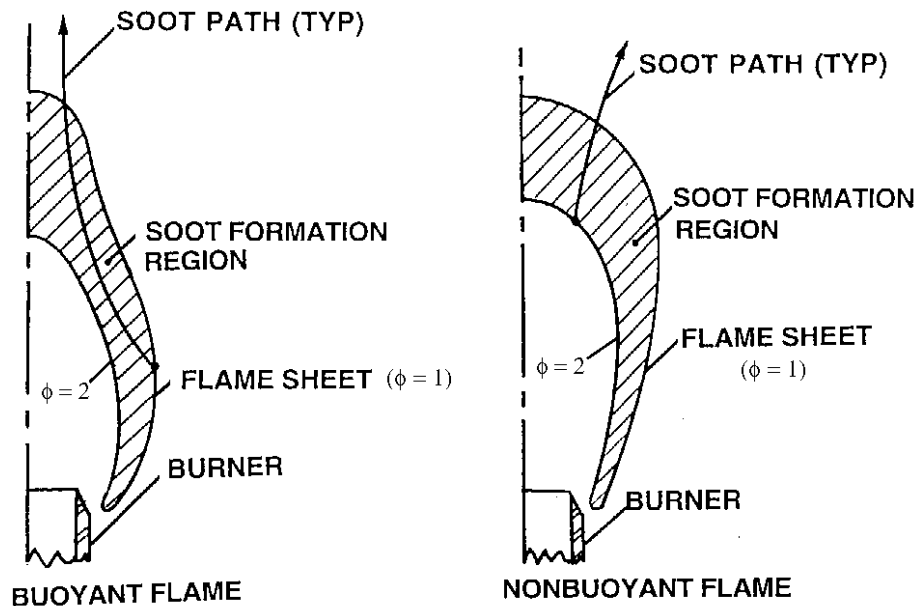


Figure 1. Soot path lines through buoyant and non-buoyant flames

The information of smoke-point properties and flame shapes is needed to improve our understanding of heat and mass transfer mechanisms for different flame conditions. Unfortunately, while the laminar smoke-point properties of buoyant jet diffusion flames at normal gravity are well known, corresponding information for nonbuoyant flames is limited and questionable. Thus the insights of soot generation and oxidation processes under microgravity conditions are critical to devise better fire safety strategy in spacecraft needed for long term space travels and other space exploration activities. In addition, the improved understanding of soot processes in laminar flames will help us, by analogy, to gain better understanding of soot processes in practical turbulent flames.

The objectives of the SPICE experiments is to find the laminar smoke-point properties of non-buoyant jet diffusion flames for several fuels under different nozzle diameter/co-flow velocity configurations. Luminous flame shape measurements would also be made to verify models of the

flame shapes under co-flow conditions. In particular, the experiments are planned to be conducted under the following conditions: six types of fuels: methane, ethylene, propane, propylene, 75% propylene mixed with 25% nitrogen in moles, and 50% propylene mixed with 50% nitrogen in moles; three nozzles: 0.4, 0.8 and 1.6 mm ID and two air velocities: 100% and 50% of the maximum air flow velocity (near 50cm/s). The operation procedures are as such that the air velocity is set first, then start the fuel flow and ignite the flame. Fuel flow rate is adjusted until the smoke point is reached. Thus, for each combination of a fuel and a nozzle size, two smoke point fuel flow rates are obtained. One is under 100% air velocity and the other is under 50% air velocity. Then in the case of 50% air velocity, the fuel flow rate is reduced to 50% of the smoke point fuel flow rate to get a flame length with that fuel flow rate.

SPICE EXPERIMENT SETUP

A. The SPICE Apparatus

The SPICE project utilized and modified an apparatus constructed for a previous project, Enclosed Laminar Flames (ELF), which was conducted in the Middle Deck Glovebox of Space Shuttle Columbia during mission USMP-4/STS-87 in November, 1997. The SPICE apparatus configuration and the connection to the MSG power supply are shown in Figure 2. The upper right portion in the figure, separated by dashed lines includes the parts placed inside the Glovebox, whereas the rest parts are outside the Glovebox, either for astronauts to control or monitor the experiment inside.

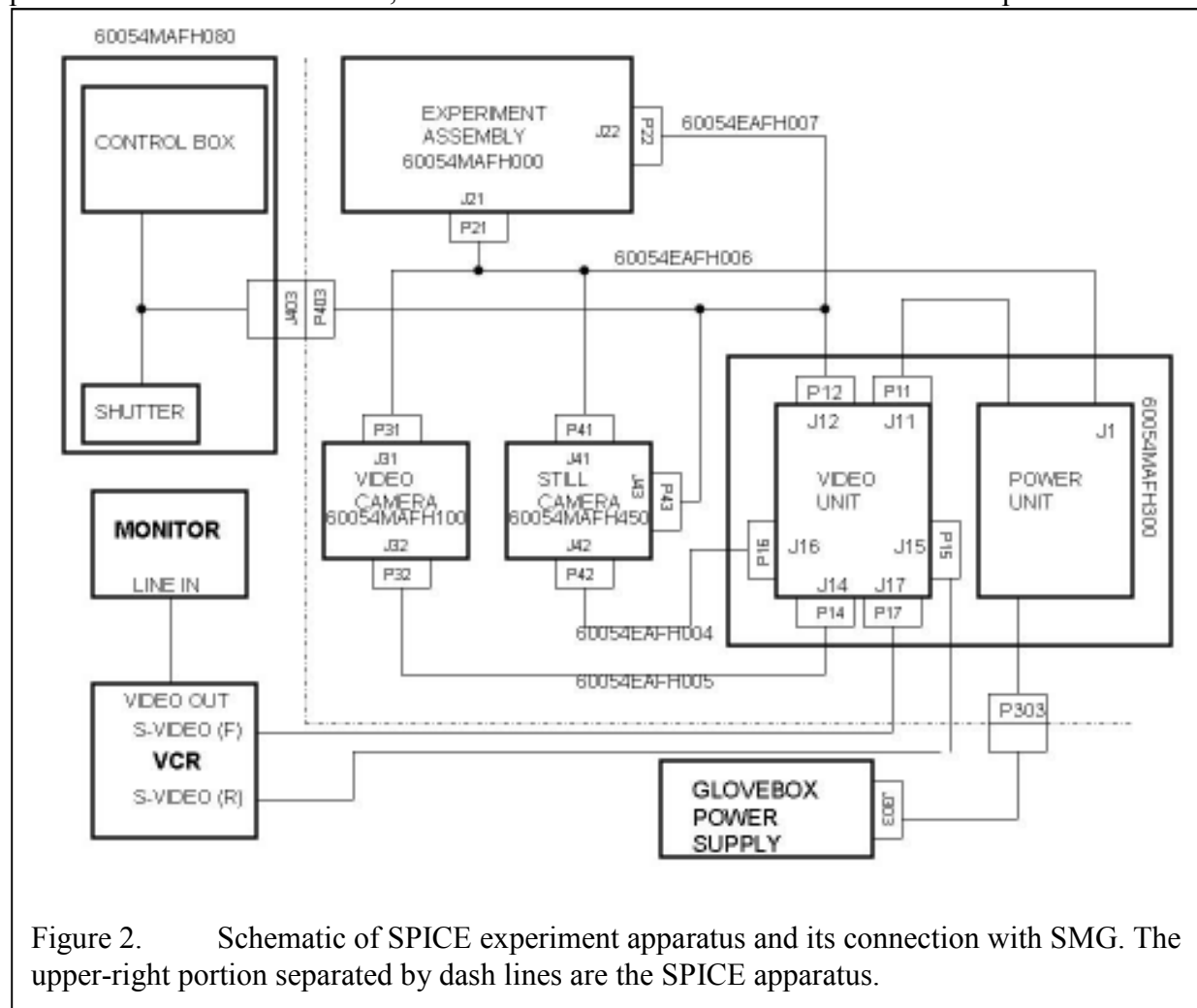


Figure 2. Schematic of SPICE experiment apparatus and its connection with SMG. The upper-right portion separated by dash lines are the SPICE apparatus.

Inside the Glovebox, the core of the apparatus is the Experimental Assembly which consists of a flow duct and its accessories. In this flow duct burns the flame under investigation. More detailed discuss of this part will be given below. An imaging system, including a still digital camera and a video camera is mounted outside the flow duct to record the flame images. Inside the Glovbox there are two additional enclosures. One is the Power Unit that converts the electrical power provided by the Glovebox Power Supply (single voltage) to various voltages needed by different part of the SPICE apparatus. The second enclosure is the Video Unit which has the following functions: a) controls the shutter of the still cameras based on the signal received from the control box, and b) merges the video camera signal with the signal from a title generator which includes air and fuel flow rates and outputs the merged signal to a VCR outside the Glovebox for recording and displaying on a monitor. The feature of merging flame image and the flow rates to a single video image is very important improvement of SPICE over ELF. With the real time flow rates stamped on the flame image, the data reduction process, after the experiment, is greatly simplified by eliminating the need of synchronizing the flame image and the flow data manually, which would be very tedious and time consuming.

As shown in Fig. 2, outside the Glovebox there are two parts: a control box that astronauts hold to control the experiment in the flow duct and a video recording and display part for astronauts to watch the flame during burn, because astronauts need to determine when the flame starts to emit soot and when a still image should be taken. More detailed discussion of each part is given below.

Experiment Assembly: Figure 3 shows a photo of the Experiment Assembly.



Figure 3. A photo of the SPICE apparatus. Air flows from left to right.

The flow duct is the core of the Assembly. The interior dimensions of the flow duct are 172mm x 43mm x 43mm. It has two Lexan windows on its top and front sides through which the still camera and the video camera can image the flame.

A retractable ruler 105mm in length is placed inside the flow duct. It is used to generate a reference image for flame length measurement. This ruler is normally rests at the far, bottom corner of the flow duct. When a reference image is taken, it is moved in line with the nozzle tip. Since its surface is 45 degree tilted, it can be seen both by the still camera and the video camera. After the reference

image is taken, the ruler is moved back to its home position. The ruler is mechanically linked to and controlled by a handle outside of the flow duct, at its front right corner.

A hot wire igniter, a coiled Kanthal wire, was used to ignite the flame. During ignition, it is moved to the nozzle tip through a handle next to the ruler handle and electrically powered up with about 4 amps of electric current that makes the wire bright red hot. Once the flame ignition is confirmed, the igniter is then moved away from the nozzle tip to its home position. The two handles are designed in such that only one of the two, the igniter and the ruler, can be moved to the nozzle at a certain time. This provides a safety feature of the system that prevents the ruler from being burned by the igniter. There is a small magnet mounted on the linkage to the igniter and a Hall-effect sensor mounted on the hinge of the front door. This sensor controls the igniter power circuits that when the front door is in open position or when the igniter is at its home position, the igniter power circuit is disabled. In other words, the igniter can be powered only if the front door is closed and the igniter is at the nozzle tip. The igniter is replaceable. When replacing the igniter, the astronaut must open the front door so that the igniter will not be powered during the replacement.

There are 3 fuel nozzles made for the SPICE experiment. They all have the same length; 50 mm, with different inner diameters: 0.41 mm, 0.76 mm and 1.6 mm, respectively. Only one of the three nozzles is installed during a burn. Figure 4 is a photo of the igniters and nozzles in a storage box shipped to ISS. It can be seen that a nozzle has two parts: a thicker base with female threads in it and a piece of thin tubing extended from the base. Thus the nozzle is threaded on a connector inside the flow duct, on the left hand side in Fig. 3. After installation, the nozzle is along the center line of flow duct. Its tip is about 54 mm from the screen at the left end of the flow duct.

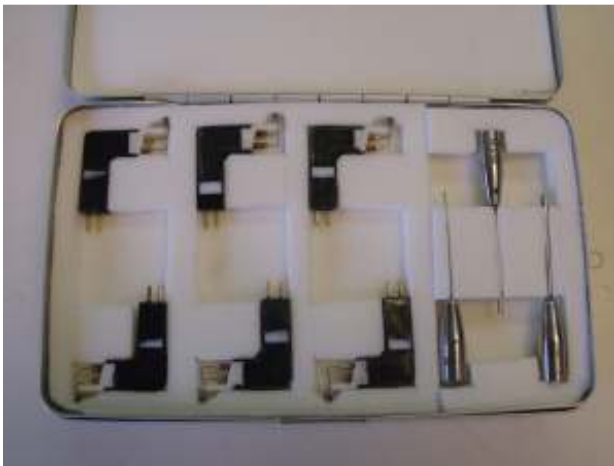
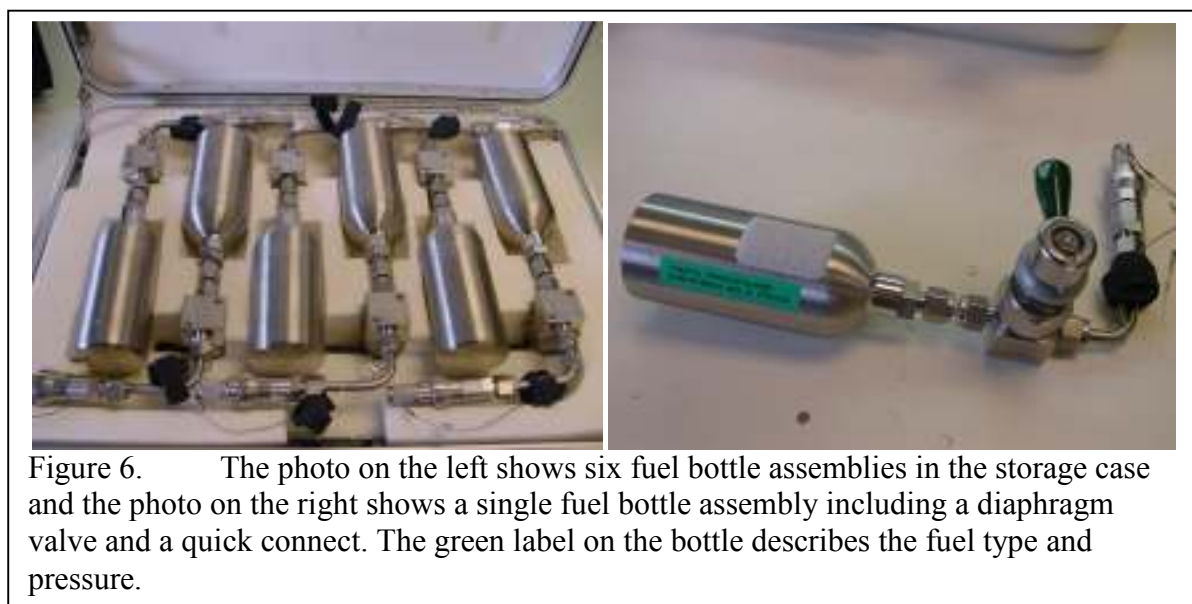
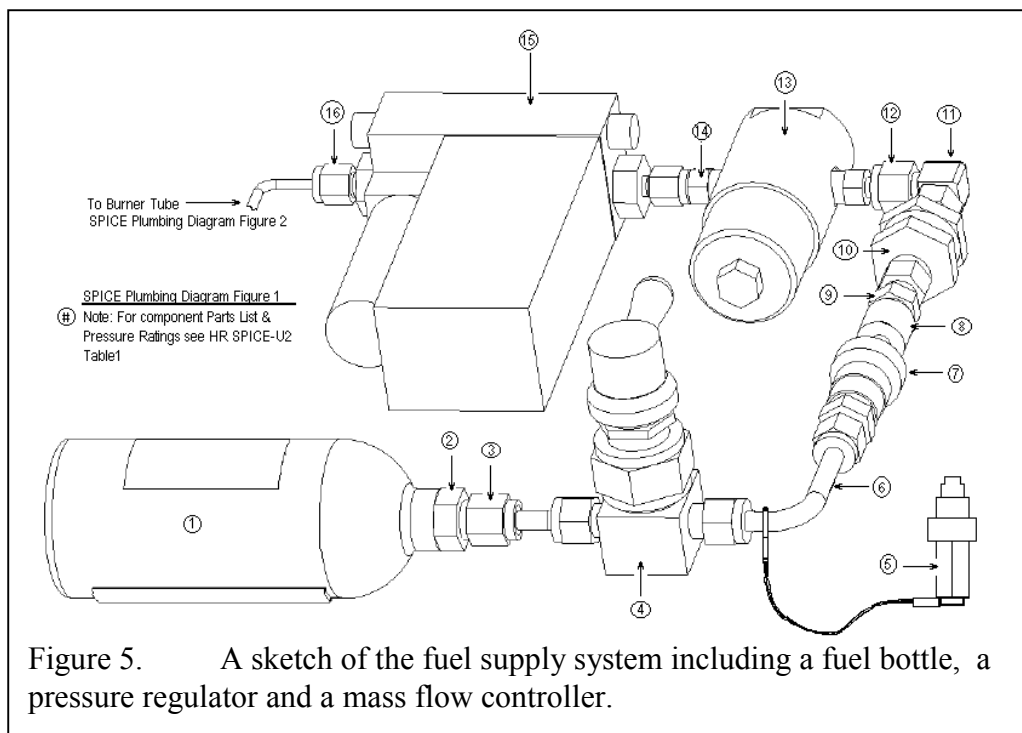


Figure 4. This photo shows three nozzles and six igniters are stored in the storage case for being uploaded to the ISS.

A thermopile radiometer, Oriel Model 71768, is placed in the far right corner downstream of the duct. It has a 90 degree view angle that can bracket the entire flame length. Since the radiometer signal is an integrated result of flame radiation, which is a complicated function of flame shape and temperature distribution in the flame, it can only provide a qualitative measurement of flame radiation. The flame radiation varies in a wide range depending on flame conditions so that the radiometer amplifier is equipped with a gain control switch to ensure the signal is detected with proper accuracy.

As shown in Fig 3, there are two rectangular boxes attached to the each end of the flow duct. The box on the left hand side has a 3" diameter electrical fan, Panasonic Model FBA08A12H1A, that blows air into the flow duct. A ceramic honeycomb flow strainer is installed downstream of the fan in the fan housing to break swirls. An omni-directional anemometer, TSI model 8475-3-xx, is located inside the flow duct near the nozzle connector to measure the air velocity. Several screens are located at the inlet of the fan, downstream of the anemometer and at the exhaust end of the chamber, respectively. Note that the fan is installed upstream of the flame so that it will not be heated by the flame gases. The box at the right end of the flow duct contains a heat sink/hand guard and a fine screen that prevents excessive soot particles from entering the ISS MSG flow system.

Attached to the flow duct is a fuel supply system illustrated in Fig. 5. As marked in the figure, items 1 to 7 are the pre-assembled fuel bottle assembly each of which includes a DOT 150 ml stainless steel single end cylinder (SWAGELOK, 304L-055MS6-150), a 1/4" diaphragm valve (SWAGELOK NUPRO # SS-DLS4), a quick connect, a Lanyard® clip, and connecting tubes. There was a long storage time from the delivery of fuel bottles from GRC to the actual execution of the experiments on ISS. Thus, a very low leak rate of fuel bottle assemblies is a critical requirement. There are total 12 fuel bottle assemblies which are stored in two separate containers. The photo on the left of Figure 6 shows one of the fuel bottle containers that holds six fuel bottle assemblies and the photo on the right shows a single assembly. The six bottles in one container are filled with six different fuel gases to the maximum safety pressure for that fuel: ethylene (210 psia), methane (210 psia), propane (110 psia), propylene (135 psia), 75% propylene (180 psia), 50% propylene (180 psia). The connection between the fuel bottle assembly and the rest of the flow system is through a couple of a 1/4" Swagelok® quick-connect, items 7 and 8 in Fig. 5. Item 8 through 16 are the plumbing system included in the Experiment Assembly, underneath the flow duct in Fig. 3. Components in the plumbing include the male half of the quick-connect (item 8), a filter (item 10), a compact stainless steel pressure regulator (item 13, Process Valve, CPR1-1B01A3E178, 0-50 psig, P/N: 106153), a mass flow controller (item 15, MKS, Type 2179 MFC with a Positive Shutoff Valve, Model #: 1179A52CS1BV-S, 0 - 500 SCCM, Standard Conditions: 0°C and 760 Torr), and stainless steel tubings. The mass flow controller sets the flow rate of the fuel gas based on the signal from the control box that astronaut holds during experiment.



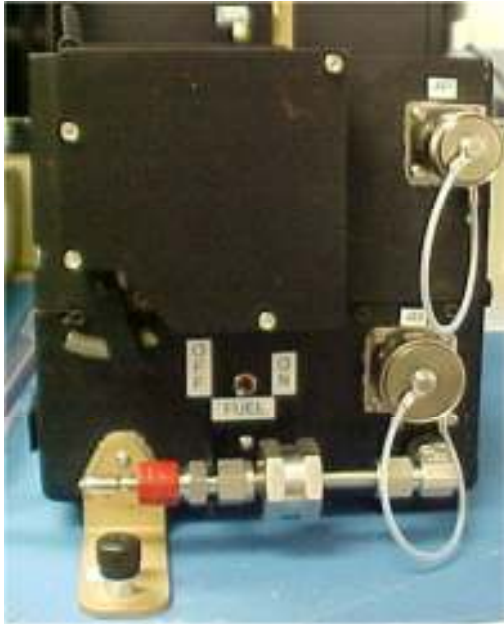


Figure 7. A side view of the SPICE apparatus shows a red connect to the fuel bottle assemblies.

The fuel bottles are exchangeable. Once a bottle is used up, it can be disconnected from the Experiment Assembly and a new bottle can be installed. Figure 7 is a side view of the Experiment Assembly showing the connector to the gas bottle, a male Swagelok® Quick Connect in red. Also seen in the photo is the toggle switch that turns the mass flow controller on and off which prevents fuel from being incidentally released to the flow duct.

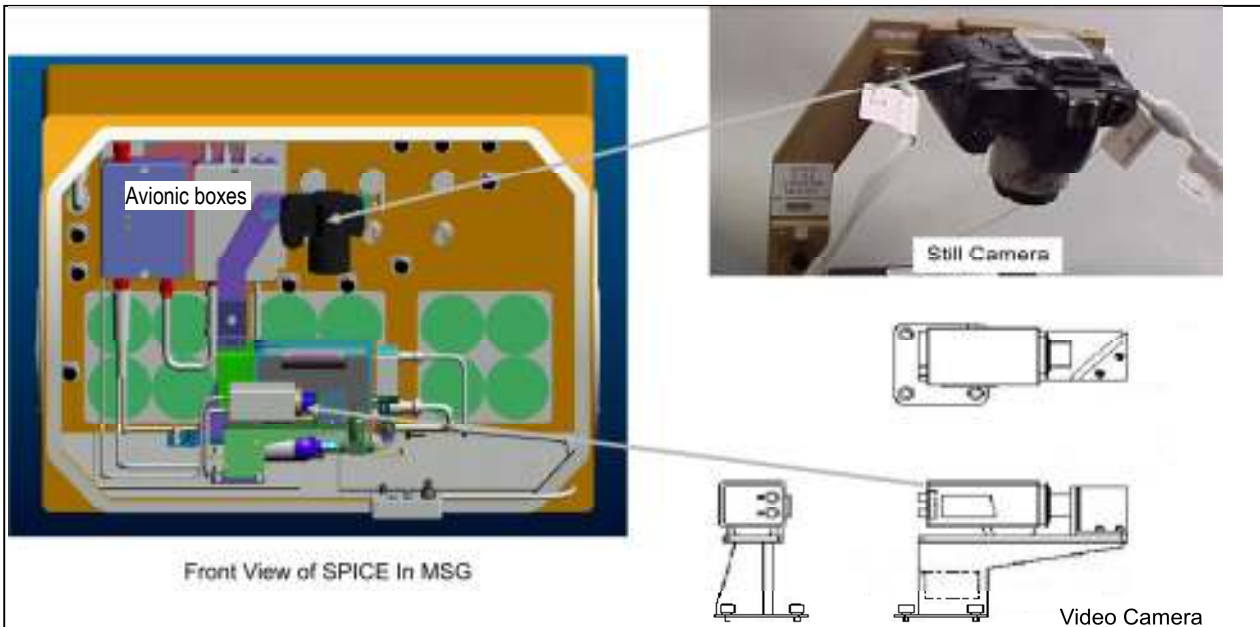


Figure 8. A front view of SPICE apparatus inside of the MSG and positions of the still camera and the video camera. Note that because of limited space, the video camera views the flame through a 45° angled mirror.

Figure 8 illustrates the installation of the SPICE apparatus inside the MSG. The still camera, a modified Nikon D100 (Fig. 9), acquires flame images from the top window of the flow duct. The video camera, Panasonic WV-CP 654 (Fig. 10), is installed in front of the flow duct providing the front view the flame. Since it is round jet flame, the top view and the front view are expected to be the same.



Figure 9. The still camera, Nikon D100, that views the flame from the top window.



Figure 10. The video camera and the mounting structure with a reflecting mirror on it.



Figure 11. The control box to control air and fuel flow rates and ignition, etc.

Figure 11 is a photo of the control box that astronaut holds in hand to manually control the experiment. There are seven controls in this control box: upper left is a 4-throw switch for setting the gain of the radiometer amplifier, which can be 10, 50, 250 or 1000. The toggle switch on the left bottom is an igniter enabling switch, which must be at “on” position for igniter to be turned on. The toggle switch next to this one is the fuel enabling switch, which must be at “on” position for fuel to flow. The toggle switch in the middle of the top row is the igniter on/off switch. The two knobs on the right are multi-turn potentiometers. The upper one controls the fuel flow rate whereas the bottom one controls the fan speed, thus the air velocity in the flow duct. On the very right side of the control box, attached is a remote shutter control of the Nikon D100 camera. When the astronaut wants to take a picture of the flame, he/she can push the button at the top of this box.

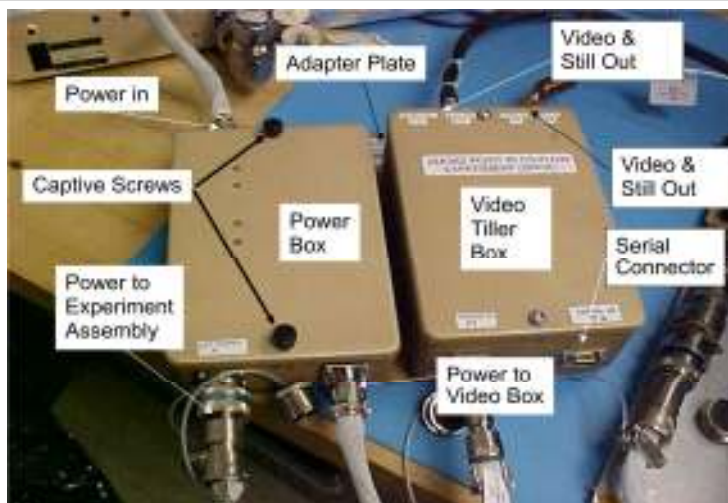


Figure 12. A photo shows the two avionics enclosures for power supply and a video titling and signal processing.

Figure 12 is a photo of the two avionic box used in SPICE. The one labeled as “Power Box” converts ISS electric power supply (28 VDC) to various power sources needed by different parts of the SPICE apparatus. The “Video Titler Box” includes a titler that can display fuel flow rate, fan speed, air velocity, and radiometer gain on the same video screen as the flame, and circuits to process video output signal and still camera images to digital data for downlink and data storage. The module is also equipped with data logger and PCMCIA storage.

B. Air velocity distribution in the flow duct

An important feature of SPICE is that it has air co-flow surrounding the flames. SPICE is the first experiment that studies soot processes of laminar jet diffusion flame under long term microgravity condition with air coflow. Because the velocity distribution in or near the flame is complicated by the varying gas property and thermal expansion, the coflow air velocity can only be specified in far field and the inlet of the flow duct as boundary conditions of the flame under investigation. For this reason, the information of air velocity distribution before igniting of the flame is needed. The air velocity was measured with an anemometer at different locations in the flow duct at different fan speed. The top window and the front window were replaced by two windows with holes on them to accommodate anemometer prob. Figure 13 shows various hole locations through which air velocities were measured. Some expressions used in describing a spatial location are explained as follows. A “Station” refers to a plane perpendicular to the flow direction. The value of a “Station” indicates the distance between the nozzle tip and the plane. There are only two Stations in use: 13 mm and 56 mm, defining two planes 13 mm and 56 mm downstream of the nozzle tip. “Height Off Floor” refers to the distance above the floor of the flow duct. The Floor of the flow duct is the lower wall of the flow duct, perpendicular to the front window. The three positions in Height Off Floor in use are 19 mm, 36 mm, and 57 mm define three planes in parallel with the flow duct floor. The 36 mm one is in the middle of the flow duct so that it includes the nozzle. The third direction is expressed in terms of “Centerline”, “20 mm to the Inside” and “20 mm to the Outside,” which define three planes in parallel with the front window of the flow duct. There are totally 18 intersections of these three sets of planes. Air velocities were measured at these 18 locations at 6 different fan dial settings. Note that for the safety concern, even at 0 fan dial setting, there is a minimum DC voltage applied to the fan to maintain a low velocity air flow which prevents fuel gas from accumulating in the flow duct.

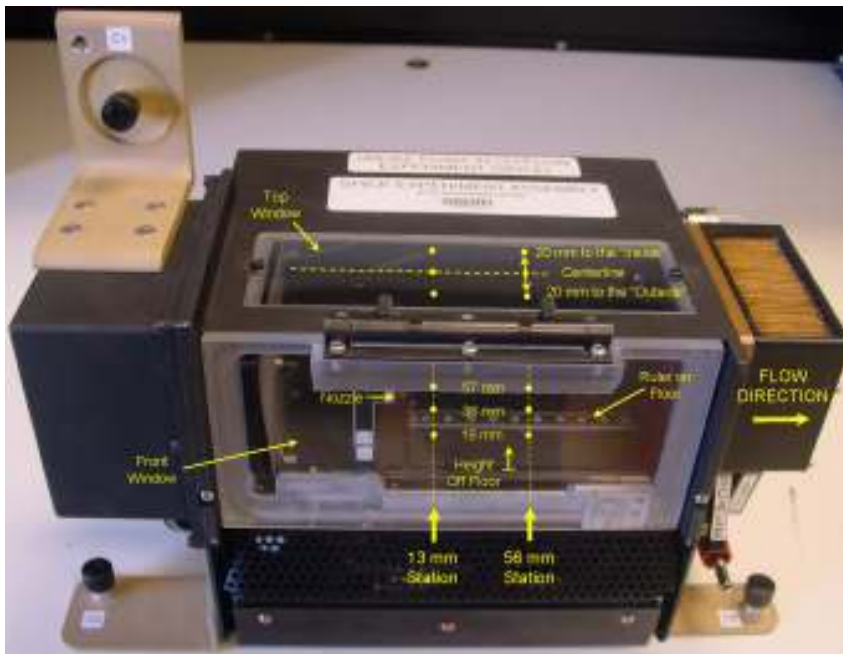


Figure 13. A photo of the SPICE apparatus with flow measurement locations marked on the top and front windows.

Four air velocity profiles are presented below in Figs. 14 and 15. Figure 14 show the air velocities on a vertical plane in parallel with the front window and at the center of the duct.

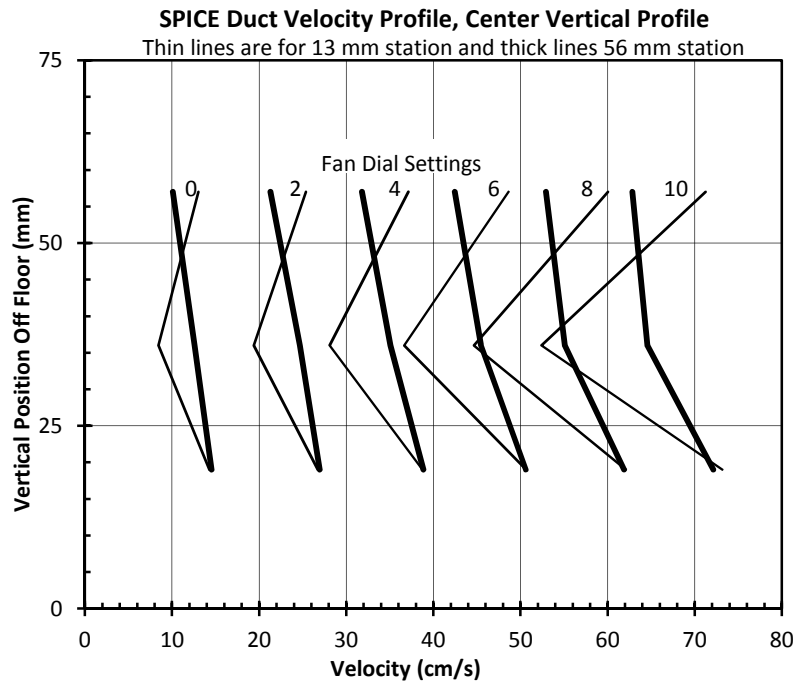


Figure 14. Air velocity distribution on the center vertical plane of the flow duct for 6 different fan settings.

The measurement was carried out by inserting the probe into the two holes, 13 mm Station and 56 mm Station, on the centerline of the top window. And three Height Off Floor locations were used in each hole. At each measuring point, 6 Fan Dial Settings on the control box were selected. It can be seen that at the 13 mm Station, the air velocity on the centerline (downstream of the nozzle) is lower than those above and below.

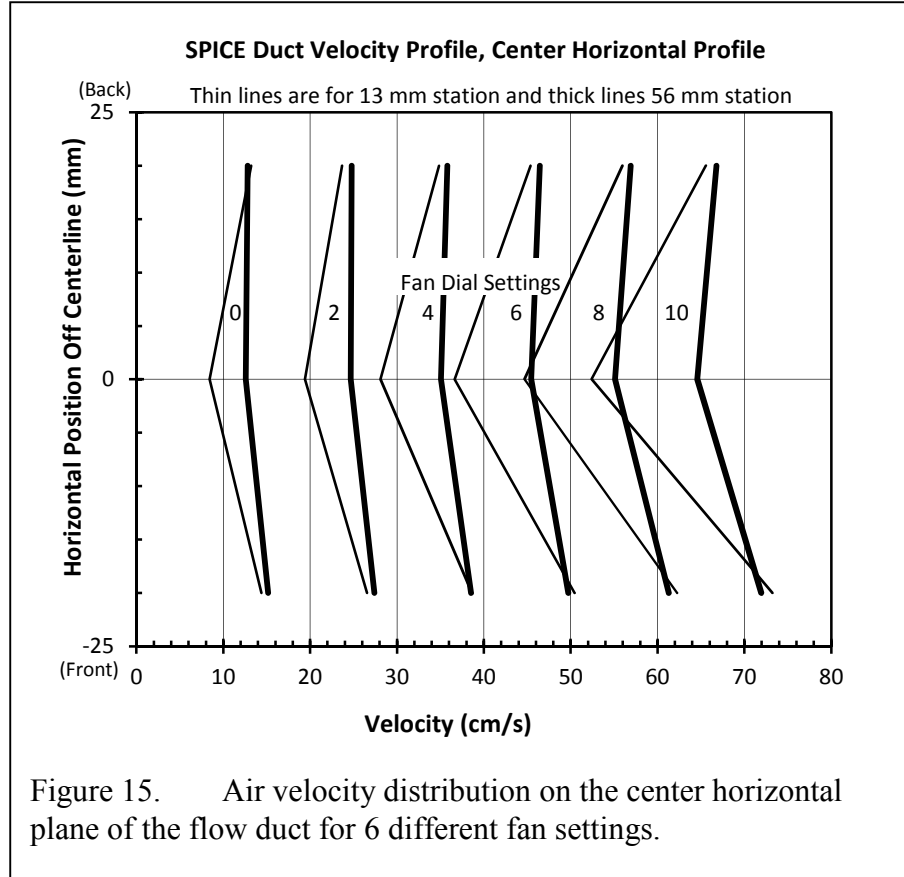
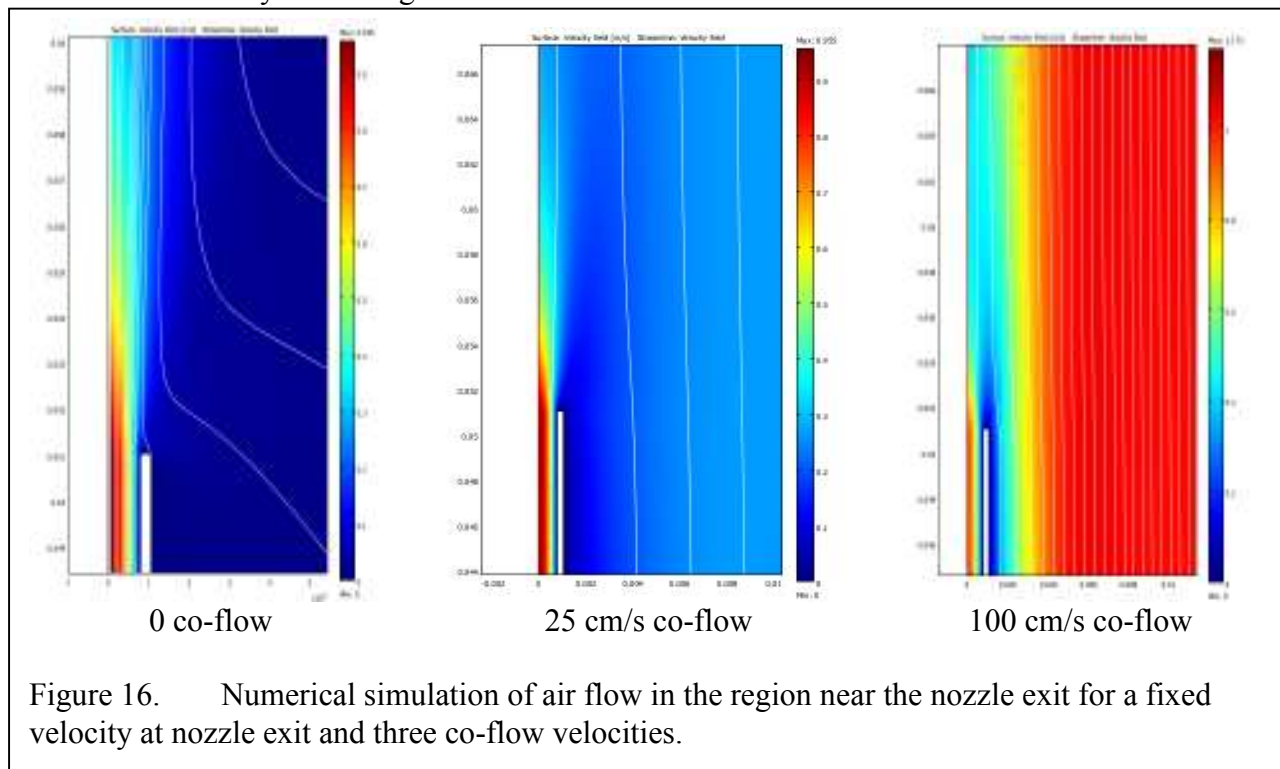


Figure 15. Air velocity distribution on the center horizontal plane of the flow duct for 6 different fan settings.

The velocity difference increases with fan settings (average air velocity), whereas at the 56 mm Station, this center-dipping velocity profile evolves to a near linear profile with higher velocity at the bottom. This center-dipping is caused by the boundary layer around the nozzle. Figure 15 shows the air velocity distribution in a horizontal plane at the center of the flow duct. The phenomenon of lower velocity along the center line can be seen in this plane too. Different from the velocity profile on the vertical plane, the center-dipping also shows at 56 mm Station.

The flow field was further modeled with COMSOL, a commercial software. The air flow near the nozzle exit was shown in Fig. 16 for three different combinations of the nozzle velocity and co-flow velocity. The white lines in the plots are streamlines. The colors show the velocity map. Blue is for low velocity whereas red indicates high velocity. Although in each plot, velocity varies from blue to red, the absolute value of certain color is different between these velocity maps. The nozzle ID in all the three cases is 1.6 mm and the average air velocity in the nozzle remains to be 50 cm/s whereas the coflow velocities are 0, 25 and 100 cm/s, respectively as stated below each plot. It can be seen that for the case of zero co-flow, the lateral air velocity profile becomes monotonically decay with its maximum value being at the center line. When the coflow velocity increases to 25 cm/s, a boundary layer around the nozzle becomes significant. Near the nozzle, in the radial direction the air velocity decreases with radius and then increases. As radius further increases to get near to the flow duct wall, the air velocity drops again. This type of velocity profile extends to quite large distance downstream of the nozzle. When coflow velocity increases to 100 cm/s, the coflow is significantly

faster than jet flow. Under this condition, the area affected by the nozzle boundary layer becomes small. In relative short distance downstream the nozzle, the velocity profile in radial direction becomes monotonically increasing with radius.



Thus the actual velocity distribution in the flow duct is quite complicated. It not only depends on the jet velocity and the coflow velocity, but also is affected by the heat released from the flame. The above discussion gives us a qualitative insight of the velocity field. Limited by the space in the MSG, SPICE experiment does not have velocity diagnostic capability to reveal the actual velocity in the presence of flame.

SPICE FLIGHT TESTS

The actual SPICE tests were conducted in SMG in twelve sessions from February 14, 2009 to June 23, 2009. In each session, the SPICE science team and engineering team were in the TeleScience Center (TSC) in GRC to support the tests conducted on MSG. Real-time flame images were available as long as air-to-ground communication channel was available. In addition, the team was able to communicate with the astronaut who conducted the experiment to answer questions or to clarify any ambiguities.

Most tests were conducted by setting the coflow velocity, igniting the flame, and adjusting the fuel flow rate. Approximate smoke points were identified in the video by the flight crew and simultaneously by the ground support team in TSC. Typical burn times were 60 s, and after each two such burns the MSG contents were flushed with ISS cabin air. The oxygen mole fraction of this cabin air is maintained within 19 – 24%.

After the tests the video record was analyzed to identify smoke points. The smoke point lengths, L_{SP} , (and the flow conditions) were obtained from the video record and are the measured distance from the burner to the tip of the luminous flame on the flame axis. The luminous flame tips were not sharply defined in most images, but rather transitioned from bright yellow to black over an axial distance of 2 – 5 mm. Luminous lengths were measured to the point where the centerline grayscale

intensity changed most rapidly between these yellow and black regions. This helped reduce the sensitivity of the measured lengths to camera exposures.

Totally 526 flames were ignited and studied, which were much more than originally planned. The reason more flames can be studied was owing to the increasing familiarization for astronauts to the apparatus, so that the flame adjusting time and smoke point realization time became shorter and shorter. This leads to plenty unused fuel for new test points. The entire test matrix and measured flame data are listed in Appendix A.

The flames generally reached steady conditions within 5 s of ignition or a change in flow rate of fuel or air. Flames of propylene and propylene mixtures were generally more luminous than the others, attributed to increased soot volume fractions. Periodic flame motion was observed in some flames, especially those with high air velocity, large burners, and long flames, and is attributed to coflow perturbations. This increased slightly the uncertainties in smoke points for these conditions.

TEST RESULTS

Smoke points were attempted for all fuels. The conditions for which smoke points were found are summarized in Table 1. In some cases (e.g., tests involving ethylene) the fuel supply was depleted before all desired conditions could be tested, as the supply was limited to two 150 cc bottles of each fuel at gage pressures of 760 – 1440 kPa. Smoke points were attempted with methane, but could not be obtained before the flames impinged on the copper plate at the end of the duct.

Table 1. Summary of smoke points observed^a

Parameter	C ₂ H ₄	C ₃ H ₈	C ₃ H ₆	75% C ₃ H ₆	50% C ₃ H ₆
Z_{st}	0.0637	0.0603	0.0637	0.0768	0.1018
T_{ad} , K	2367	2264	2332	2317	2288
d , mm	0.76 – 1.6	0.76 – 1.6	0.41 – 1.6	0.76 – 1.6	0.76 – 1.6
u_a , cm/s	18 – 58	16 – 45	5.4 – 57	13 – 65	11 – 49
L_{SP} , mm	37 – 105	46 – 105	13 – 59	15 – 44	22 – 50
u_f , cm/s	69.6 – 734	48.9 – 460	11.2 – 763	19.9 – 272	49.6 – 112
$\dot{m}_f$, mg/s	1.62 – 3.89	1.80 – 3.84	0.39 – 1.73	0.64 – 1.98	1.45 – 3.29
Re	127 – 636	176 – 788	36.4 – 630	49.8 – 324	89.4 – 207
# smoke points	6	6	20	13	7

^aAll smoke point flames were attached to the burner, i.e., not lifted. For mixtures the fuel mole fraction is shown and the inert is N₂. Parameters u_a , u_f , and Re assume conditions of 298 K, 101 kPa and 21±1% O₂, and uniform velocity profiles. Viscosities used to calculate Re involve burner gas properties at 298 K and 101 kPa.

Some smoke points (particularly for propane and ethylene) were identified by the onset of gradual dimming, reddening, and rounding of the luminous flame tip. Conversely, the brightest flames (particularly for propylene and propylene mixtures) normally did not display significant dimming and reddening near their tips, except when much longer than their smoke points. For these flames, smoke points were identified by the rapid transition to open-tipped flames. Open-tipped flames are common in microgravity when smoke points are far exceeded⁴ and are associated with flame extinguishment along the centerline and soot that escapes in an annular shell. For conditions sufficiently above the smoke points, quenched soot could be seen leaving the flames despite the dark background in the video.

Figure 17 shows images for representative flames near their smoke points. Two sequences are shown, and in both, the smoke point occurs between the third and fourth images. In the top row, coflow velocity decreases from left to right at a constant fuel flow rate. This increases luminous flame length and leads to soot emission in the fourth image. The luminous length decreases in the fourth image owing to flame extinction and extensive reactant leakage. In the bottom row, fuel flow rate is increased at a constant coflow velocity, increasing luminous lengths. The luminous lengths of the flames in Fig. 17 are more sensitive to changes in fuel flow rate than to changes in coflow velocity. For a jet diffusion flame, the fuel flow rate is the limiting factor of the flame length because there is almost unlimited amount of oxidant (air in this case) available. This is why the flame length is more sensitive to the change of fuel flow rate.

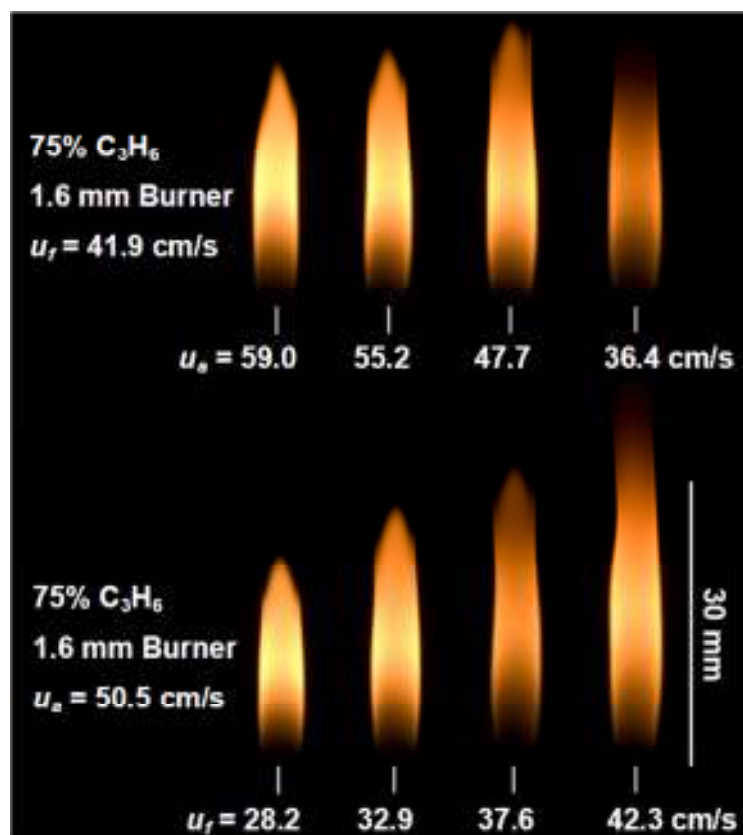


Figure 17. Still camera images of 75% C_3H_6 flames with (top) decreasing coflow velocity, and (bottom) increasing fuel flow rate. Images were taken at f/11 with a shutter speed of 3.1 ms, except the rightmost image in the top row and 3rd image in the bottom row, which used a shutter speed of 1.3 ms in original color.

In Fig. 18 the measured smoke point lengths are plotted with respect to coflow velocity. For a given choice of fuel and burner diameter, an increase in coflow velocity generally increases the smoke point length. This behavior is attributed to decreased residence times and radiative losses as coflow velocity increases, and is an important issue for further consideration. The slope of the correlations in Fig. 18 generally decreases with increased sooting propensity. Figure 18 reveals that smoke points of nonbuoyant coflowing flames decrease with increasing burner diameter. This behavior also can be attributed to effects of residence time, as fuel velocity is proportional to d^{-2} when fuel mass flow rate is constant.

Also shown in Fig. 18 are smoke points of microgravity flames in still air.⁵ The limited smoke points at different diameters actually increase with burner diameter. When the present measurements are extrapolated to zero coflow velocity, the dependence on burner diameter diminishes. Normal gravity smoke points are also generally found to have only a weak dependence on burner diameter.^{5,6} However, testing in normal gravity⁸ has shown that coflowing normal gravity smoke points for propylene decrease with increasing burner diameter for the present range of diameters. The present smoke points generally were repeatable. An exception was the tests with the 0.41 mm burner (for which only propylene smoke points were possible). Testing on different days yielded different smoke points for this burner.

The sooting propensity of the present fuels, identified from the smoke point lengths for fixed burner diameter and coflow velocity, generally increases from propane to ethylene, to 50% propylene, to 75% propylene, and to propylene. This is consistent with normal gravity results in the same apparatus.⁷ The only exception was that, for the 0.76 mm burner, ethylene has a longer smoke point length than propane. The lower density of ethylene results in a decreased residence time, but this is unlikely to explain such a long smoke point relative to that of propane.

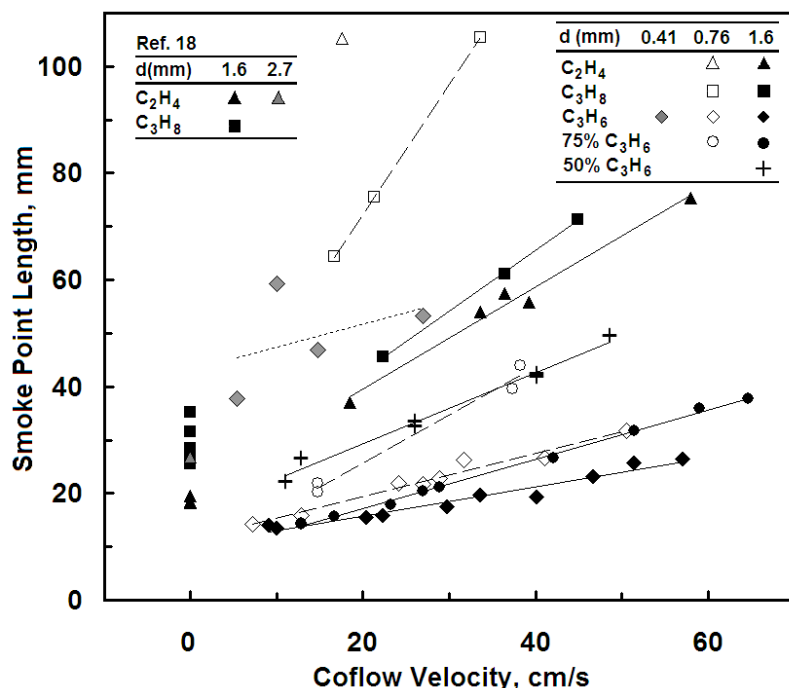


Figure 18. Smoke point luminous length plotted with respect to coflow velocity. The lines shown are fits for each pairing of fuel and burner diameter for present data.

In Fig. 19 the smoke point flame lengths are plotted with respect to $\dot{m}_f / Z_{st}$. This is motivated by Eq. (2) for stoichiometric lengths and the past observations that flames at their smoke points typically have luminous lengths that are twice as long as their stoichiometric lengths. This choice of axes yields a reasonably good single correlation of the luminous lengths of all fuels. These results are similar to those of Lin and Faeth¹ in that flames with low u_a / u_f are generally longer than those at otherwise matched conditions. This occurs in Fig. 19 for propylene, propane, and ethylene flames with the smallest burner, for which $u_a / u_f < 0.08$.

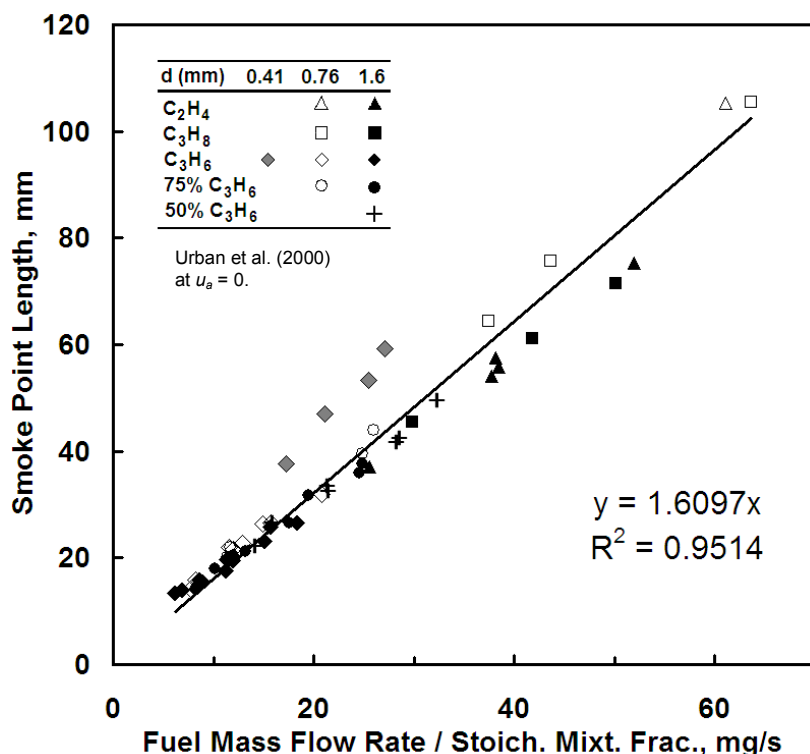


Figure 19. Smoke point luminous length plotted with respect to fuel mass flow rate divided by stoichiometric mixture fraction. A linear fit that intersects the origin is shown.

DISCUSSION

Although it is convenient to classified all flames, laminar vs. turbulent and diffusion vs. premixed, into two categories based on their soot emission conditions, i.e., soot emitting flames and non-soot emitting flames, the conditions delineating the two or the conditions of onset of soot emission are difficult to define. Up to date, these conditions have been documented solely based on experimental observations. Many efforts have been made to develop a universal method of defining the onset of soot emission, i.e., smoke point, conditions, in terms of primary parameters of the flame. But none of these efforts has been quite successful yet. The primary parameters of the flame mentioned here refer to parameters used to set the flame, not those generated by the flame. The primary parameters of jet diffusion flames include fuel type, fuel and oxidizer supply rates, ambient temperature and pressure, and configuration of the device where the flame takes place, such as nozzle diameter and dimensions of the combustion chamber. The flame temperature, gas velocity and species distributions as well as flame shapes and derived parameters such as heat generation and heat loss, are the parameters generated by the flame which is not considered as primary parameters. If a method that relates smoke point to those primary parameters were found, we would be able to predict whether a flame emits soot or not based on the planned flame conditions. In the course of pursuing such a universal method, a number of surmises of the mechanism of soot emission have been proposed to associate smoke point with certain flame generated parameters. Those surmises served as a middle step in hoping to help further development of the aforementioned method. The first surmise, and least controversial, is that smoke points occur when soot temperature drops below a critical temperature prior to soot burnout. This temperature is about 1300 K in normal gravity^{8,9} and 1000 K in microgravity.^{10,11} The second is that smoke point occurs when radiative loss fraction reaches 0.2 – 0.4 for normal-gravity flames¹²⁻¹⁶ or 0.4 – 0.6 for microgravity flames.¹⁷ Since the radiative fraction of heat loss increases with fuel flow rate, smoke point will occur as the fuel flow rate is higher than

certain value. This is consistent with experimental observations. The third surmise is that the smoke point correlates to the ratio of luminous length divided by stoichiometric length. This ratio increases with flow rate. Once it reaches about 2, smoke point occurs.^{15,18,19} The last surmise suggests that smoke point correlates with the centerline flame residence time. This is the most complex^{8-11,17,18,20-24} surmise because flame residence time affects the soot processes in twofold. An increase in residence time can lead to soot emissions by increasing the time available for soot formation. Because this generally also increases the time available for soot oxidation, it has been proposed that smoke points are associated with conditions where the time available for soot formation exceeds that for soot oxidation to certain extent.²¹ This is difficult to prove because to evaluate the flame residence time requires the velocity distribution along the centerline and the flame length, both of which are resulted flame parameters, not readily available from primary parameters of the experiment. In addition, the soot formation time and oxidation time are conceptually defined. Practically, these times are not measurable and they might be overlapped, i.e., for the same soot aggregate, adding and oxidizing carbon particles may happen simultaneously. Because of these difficulties and ambiguities, the flame residence time theory cannot be proved or disproved rigorously. However, these obstacles do not prevent us from checking the residence time theory against the observed smoke point data with the help of some approximated velocity distribution and flame length information. More detailed discussion of this effort can be found in the publications resulted from the SPICE project listed in Appendices B through F. The results showed that although residence time is helpful in understanding the present smoke points, it does not yield quantitative correlations.

CLOSING REMARKS

Smoke points of 55 microgravity gas-jet diffusion flames in coflowing air were measured in earth orbit. The conditions emphasized small burners and the effects of coflow velocity. These flames allow extensive control over residence time, which is important in smoke point phenomena. The major conclusions are as follows:

1. An increase in coflow velocity increases the smoke point length.
2. A decrease in burner diameter increases the smoke point length.
3. These two effects can be explained qualitatively by considering residence times; increased fuel and/or coflow velocities yield increased smoke point lengths.
4. The sooting propensities of the present fuel mixtures in microgravity increase according to ethylene < propane < 50% propylene < 75% propylene < propylene. This is the same as in normal gravity except for the exchanged positions of ethylene and propane.
5. Consistent with past theory and experiments, luminous length at the smoke point is proportional to fuel mass flow rate. However there is a slight increase in luminous length as u_a / u_f decreases.
6. Although residence time is helpful in understanding the present smoke points, it does not yield quantitative correlations of smoke points.
7. The present smoke point lengths scale with $d^{-0.91} u_{air}^{0.41}$. Either a decrease in diameter or an increase in coflow velocity decreases residence time and radiative loss fraction, which in turn leads to a longer smoke point.

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Appendix A. (Spice Flight Test data II.xls).....	A-1
Appendix B. (2009, Eastern State Combustion Meeting, <u>paper A</u>).....	B-1
Appendix C. (2010 AIAA Paper, <u>paper B</u>).....	C-1
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Appendix F. (2010 Fire safety J. Paper, <u>paper E</u>).....	F-1

Appendix C
Appendix D

Appendix E